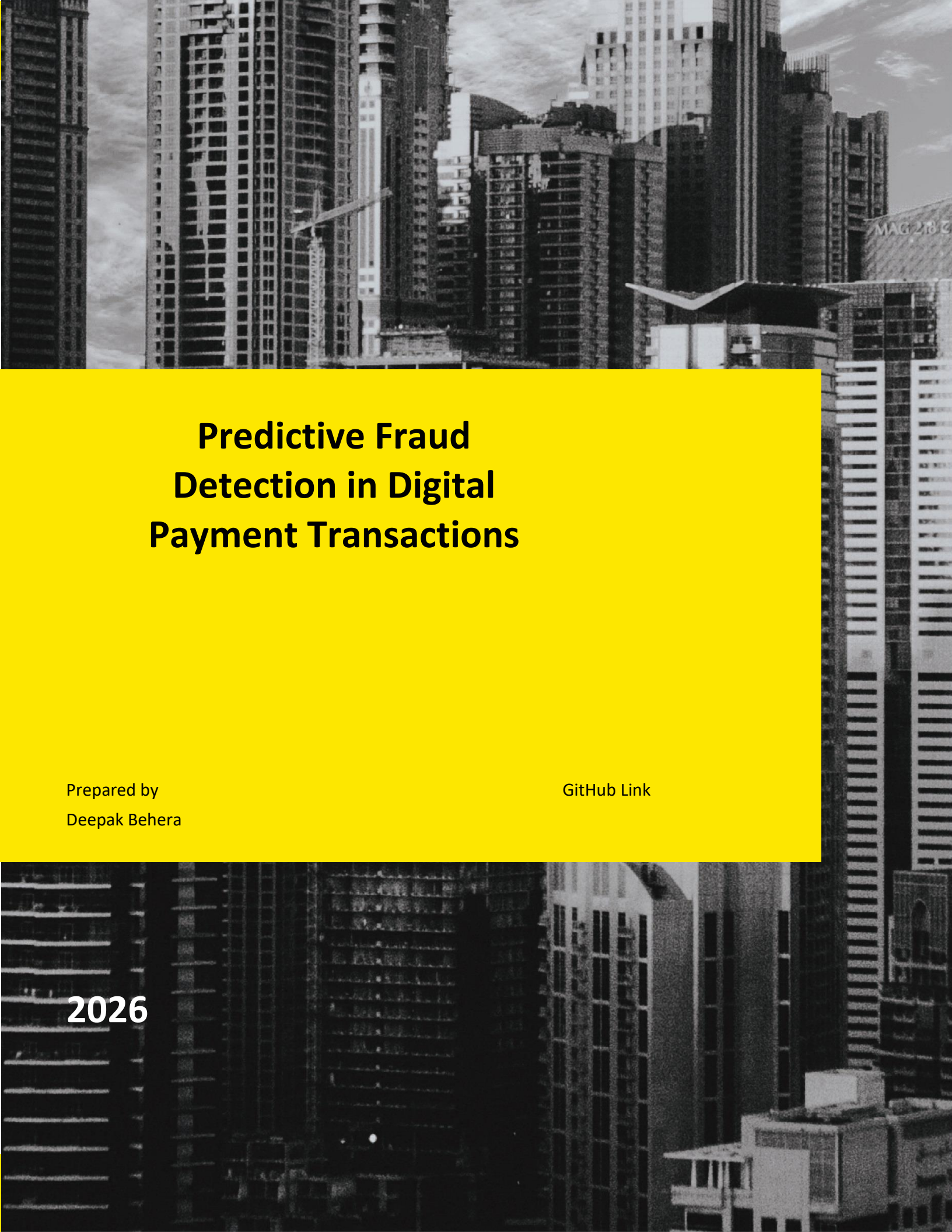




Predictive Fraud Detection in Digital Payment Transactions

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[GitHub Link](#)

2026

1. Title Page

Predictive Fraud Detection in Digital Payment Transactions



2. Executive Summary

Fraudulent transactions pose a significant challenge to retail and e-commerce businesses, resulting in financial losses and reduced customer trust. This project develops a machine learning-based fraud detection system to identify suspicious transaction patterns and predict fraudulent activities. The workflow includes data collection, exploratory data analysis, data preprocessing, feature engineering, model training, and performance evaluation using multiple classification algorithms. Models such as Logistic Regression, Decision Tree, Random Forest, Gradient Boosting, and XG Boost are compared using metrics including Accuracy, Precision, Recall, F1-Score, and ROC-AUC. The selected model provides reliable fraud predictions, enabling organizations to improve risk management, reduce fraud-related losses, and support faster, data-driven decision-making.

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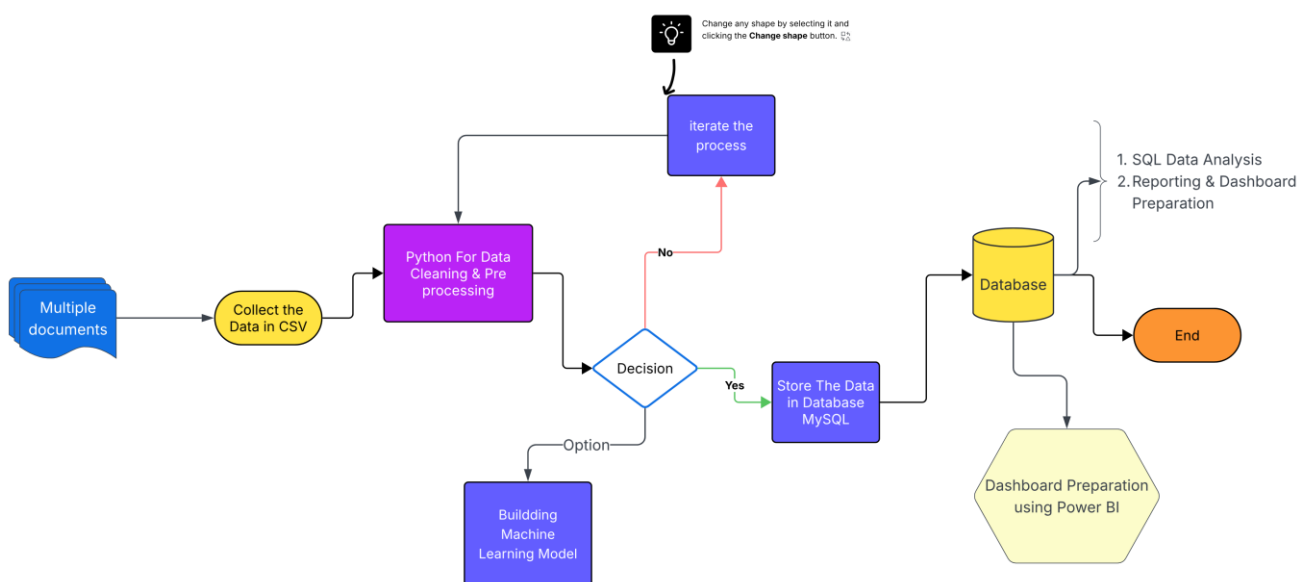
4. Introduction

Fraudulent transactions have become a significant challenge in the retail and e-commerce industry due to the rapid growth of digital payment systems. As the volume of online transactions continues to increase, organizations face a higher risk of financial losses, operational disruptions, and reduced customer trust caused by fraudulent activities. Traditional rule-based fraud detection methods often fail to identify sophisticated and evolving fraud patterns in real time. This project applies machine learning techniques to analyze transaction data, identify suspicious behavior, and predict fraudulent transactions accurately. The objective is to develop a reliable predictive model that enhances fraud detection, supports informed business decisions, and strengthens transaction security.

5. Business Understanding

Fraudulent transactions are a significant challenge for retail and e-commerce businesses, leading to financial losses, operational disruptions, and reduced customer trust. As digital payment volumes continue to grow, traditional rule-based fraud detection methods often fail to identify evolving fraud patterns in real time. This project aims to develop a machine learning-based fraud detection system that analyzes transaction data to distinguish legitimate transactions from fraudulent ones. By identifying key risk indicators such as unusual transaction amounts, high transaction frequency, suspicious locations, and risky devices, the model helps organizations detect fraud more accurately. The solution supports faster decision-making, reduces financial risk, improves customer confidence, and enhances overall business security.

6. Business Process Modelling



7. Dataset Description

<i>Attribute</i>	<i>Description</i>
Dataset Name	Retail Transaction Fraud Detection Dataset
Domain	Retail / E-commerce / Financial Services
Purpose	Detect fraudulent transactions using machine learning
Type	Structured (Tabular) Dataset
Target Variable	fraud flag (0 = Genuine Transaction, 1 = Fraudulent Transaction)
Records	100,000 Transactions
Features	17 Predictor Variables + 1 Target Variable
Data Format	CSV
Problem Type	Binary Classification

<i>Feature</i>	<i>Description</i>
<i>transaction_id</i>	Unique identifier for each transaction
<i>customer_id</i>	Unique identifier for each customer
<i>transaction_timestamp</i>	Date and time when the transaction occurred
<i>transaction_amount</i>	Total amount of the transaction
<i>payment_method</i>	Payment mode (Credit Card, Debit Card, UPI, Wallet, etc.)
<i>device_type</i>	Device used for the transaction (Mobile, Desktop, Tablet)
<i>location</i>	Geographic location of the transaction
<i>merchant_category</i>	Type of merchant (Electronics, Grocery, Fashion, etc.)
<i>is_international</i>	Indicates whether the transaction is international
<i>transaction_frequency_24h</i>	Number of transactions made by the customer within the last 24 hours
<i>avg_transaction_amount_7d</i>	Customer's average transaction amount over the past 7 days
<i>failed_transaction_count_24h</i>	Number of failed transaction attempts in the last 24 hours

<i>account_age_days</i>	Number of days since the customer account was created
<i>previous_fraud_flag</i>	Indicates whether the customer has a history of fraud
<i>unusual_amount_flag</i>	Flags transactions with unusually high amounts
<i>unusual_location_flag</i>	Flags transactions made from unexpected locations
<i>multiple_transactions_short_time</i>	Indicates multiple transactions in a short period
<i>high_risk_device_flag</i>	Flags transactions made from high-risk devices
<i>velocity_flag</i>	Indicates unusually rapid transaction activity
<i>fraud_flag</i>	Target variable: 0 = Legitimate, 1 = Fraudulent

```
[34]: retail.head()
```

	transaction_id	customer_id	transaction_timestamp	transaction_amount	payment_method	device_type	location	merchant_category	is_international	transaction_fraud_flag
0	T0000001	C20953	2026-03-04 19:56:23.679173	56.31	Debit Card	Tablet	Canada	Fashion	0	
1	T0000002	C24133	2026-03-09 19:56:23.679330	20.35	Credit Card	Tablet	India	Electronics	0	
2	T0000003	C07165	2026-01-01 19:56:23.679362	48.72	Debit Card	Tablet	UK	Travel	0	
3	T0000004	C19310	2025-12-09 19:56:23.679385	153.62	Debit Card	Mobile	Australia	Fashion	0	
4	T0000005	C25019	2025-11-09 19:56:23.679409	115.32	PayPal	Mobile	India	Gaming	0	

5 rows × 21 columns

```
[35]: retail.tail()
```

	transaction_id	customer_id	transaction_timestamp	transaction_amount	payment_method	device_type	location	merchant_category	is_international	transaction_fraud_flag
99995	T0099996	C06953	2026-01-01 19:56:25.768931	79.72	Credit Card	Mobile	Australia	Travel	0	
99996	T0099997	C06691	2026-01-25 19:56:25.768949	8.09	PayPal	Desktop	Australia	Luxury	0	
99997	T0099998	C16351	2025-09-24 19:56:25.768970	94.69	Debit Card	Desktop	Australia	Groceries	1	
99998	T0099999	C22185	2026-03-06 19:56:25.768990	63.31	Google Pay	Tablet	Australia	Travel	0	
99999	T0100000	C08043	2025-10-24 19:56:25.769007	37.45	Apple Pay	Desktop	Germany	Travel	1	

5 rows × 21 columns

```
[191]: retail.shape
# There is 10,00,000 Rows and 21 Columns

[191]: (100000, 21)

[193]: retail.columns.tolist()

[193]: ['transaction_id',
'customer_id',
'transaction_timestamp',
'transaction_amount',
'payment_method',
'device_type',
'location',
'merchant_category',
'is_international',
'transaction_frequency_24h',
'avg_transaction_amount_7d',
'failed_transaction_count_24h',
'account_age_days',
'previous_fraud_flag',
'unusual_amount_flag',
'unusual_location_flag',
'multiple_transactions_short_time',
'high_risk_device_flag',
'veLOCITY_flag',
'fraud_flag',
'fraud_risk']
```

8. Technology Stack

<i>Category</i>	<i>Technology</i>	<i>Purpose</i>
<i>Programming Language</i>	Python	Data analysis, pre-processing, model development
<i>Development Environment</i>	Jupyter Notebook / VS Code	Coding, experimentation, and model development
<i>Data Manipulation</i>	Pandas	Data cleaning, transformation, and analysis
<i>Numerical Computing</i>	NumPy	Mathematical and numerical operations
<i>Data Visualization</i>	Matplotlib, Seaborn	Exploratory Data Analysis (EDA) and visualization
<i>Machine Learning</i>	Scikit-learn	Model training, evaluation, pre-processing, and metrics

<i>Classification Models</i>	Logistic Regression, Decision Tree, Random Forest, Gradient Boosting, XG Boost	Fraud prediction and model comparison
<i>Model Persistence</i>	Joblib	Saving and loading the trained model
<i>Web Application</i>	Streamlit	Deploying the machine learning model as a web application
<i>Version Control</i>	Git & GitHub	Source code management and project hosting

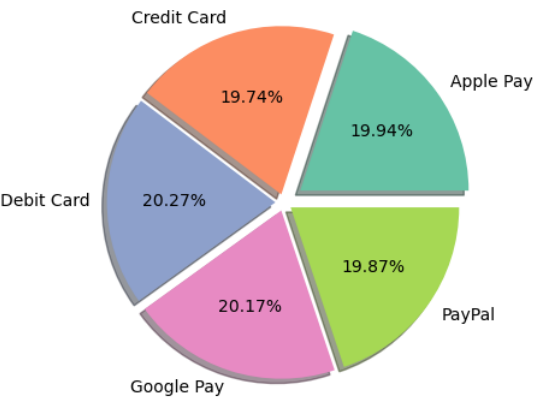
9. Data Understanding (EDA)

EDA Component	Short Description (≈20 words)	
Dataset Overview	Understand the dataset structure, number of records, features, data types, and target variable before performing further analysis.	
Data Shape	Identify the total number of rows and columns to understand dataset size and overall complexity.	
Data Types	Examine numerical and categorical variables to ensure appropriate pre-processing, visualization, and machine learning techniques are applied.	
Missing Values	Detect missing values and determine suitable handling methods to improve data quality and model performance.	
Duplicate Records	Identify and remove duplicate entries to prevent biased analysis and improve dataset accuracy and consistency.	
Statistical Summary	Review descriptive statistics including mean, median, standard deviation, minimum, and maximum values to understand feature distributions.	
Target Variable Distribution	Analyse fraud and non-fraud transaction counts to identify class imbalance and guide model selection strategies.	
Univariate Analysis	Explore each feature individually to understand distributions, variability, and potential anomalies within the dataset.	
Bivariate Analysis	Study relationships between two variables to identify trends, correlations, and fraud-related behavioural patterns.	
Correlation Analysis	Measure relationships between numerical variables to identify highly correlated features and reduce multicollinearity problems.	
Outlier Detection	Detect abnormal observations that may represent fraudulent transactions or data errors affecting model performance.	

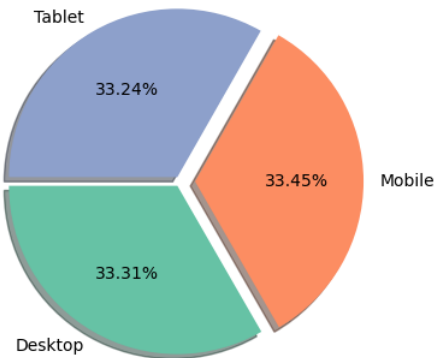
Categorical Feature Analysis	Examine category frequencies and their relationship with fraud occurrences to identify meaningful transaction behaviour patterns.
Numerical Feature Analysis	Analyse numerical variables through statistical measures and visualizations to understand trends and value distributions.
Feature Distribution	Visualize data distributions to detect skewness, normality, and transformations required before model development.
Class Imbalance Analysis	Evaluate imbalance between fraud and legitimate transactions to determine whether resampling techniques are necessary.
EDA Conclusion	Summarize key findings, important patterns, anomalies, and insights that support effective feature engineering and machine learning.



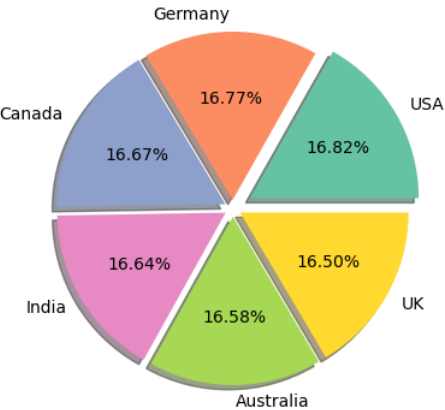
Revenue Contribution By Payment Method



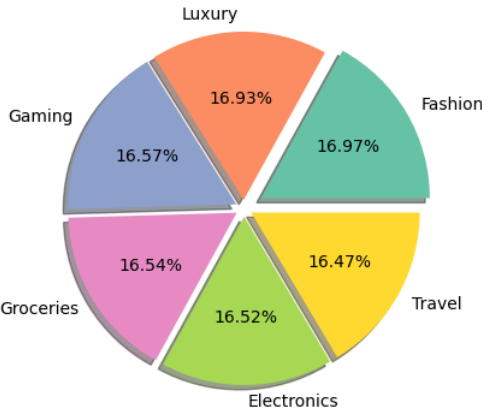
Revenue Contribution By Device Type



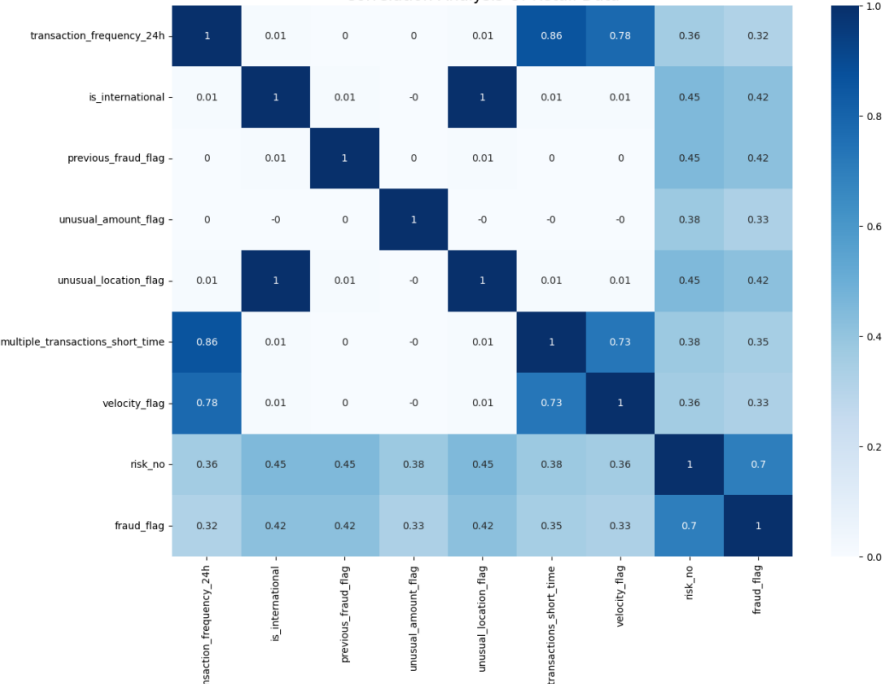
Revenue Contribution By Location Wise



Revenue Contribution By Merchant Category



Correlation Analysis Of Retail Data



10. Data Preprocessing

<i>Step</i>	<i>Short Description (~20 words)</i>
Data Cleaning	Removed inconsistent records, corrected data errors, handled missing values, and eliminated duplicate entries to improve overall dataset quality.
Missing Value Treatment	Identified missing values and handled them using suitable imputation methods or removed incomplete records for reliable analysis.
Duplicate Removal	Detected and removed duplicate transaction records to prevent bias and ensure accurate machine learning model performance.
Outlier Detection	Identified unusual transaction values using statistical techniques to minimize their impact on model accuracy and prediction reliability.
Feature Engineering	Created meaningful features from existing variables to improve predictive performance and capture hidden transaction behaviour patterns effectively.
Categorical Encoding	Converted categorical variables into numerical format using encoding techniques, enabling machine learning algorithms to process the data efficiently.
Feature Scaling	Standardized numerical features using scaling methods to ensure equal contribution and improve model training efficiency and convergence speed.
Train-Test Split	Divided the dataset into training and testing subsets to evaluate model performance on unseen transaction data objectively.

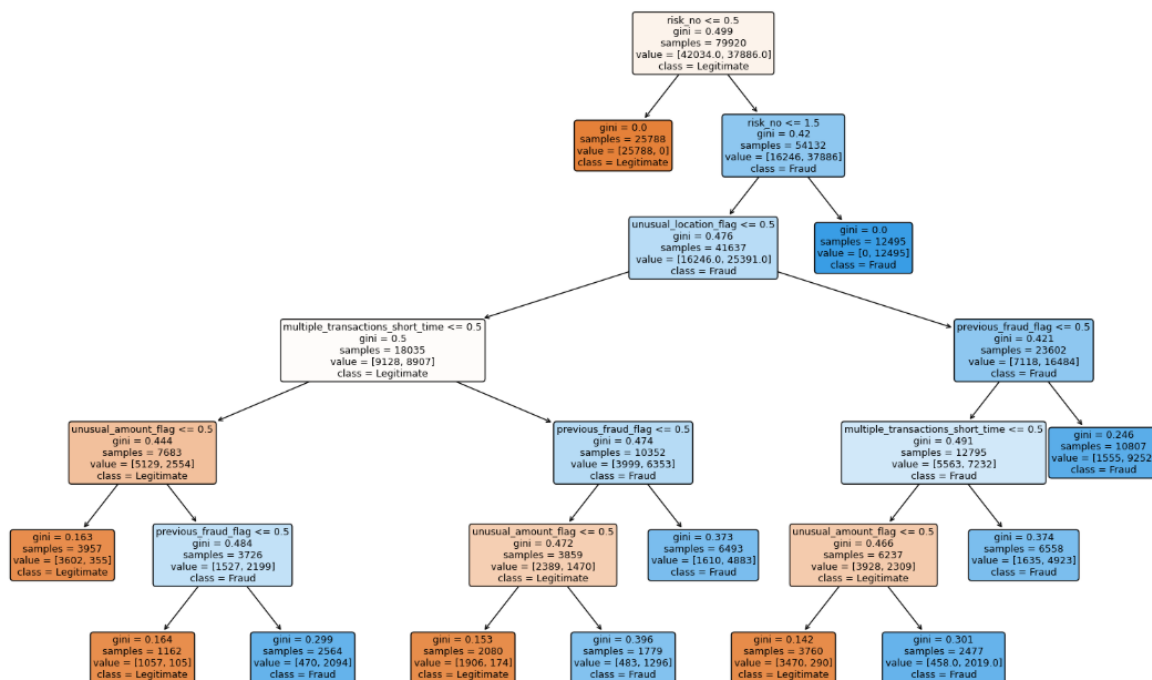
11. Feature Selection

<i>Feature Scaling Technique</i>	<i>Short Description (~20 words)</i>
Feature Scaling	Feature scaling transforms numerical values to a common range, improving model performance, convergence speed, and preventing dominance by larger features.
Standardization (Z-Score Scaling)	Standardization rescales features to have zero mean and unit variance, making data suitable for many machine learning algorithms.
Normalization (Min-Max Scaling)	Normalization scales feature values between 0 and 1, preserving relationships while ensuring consistent feature ranges across variables.
Why Feature Scaling?	Feature scaling improves model accuracy, accelerates training, ensures fair feature contribution, and enhances distance-based algorithm performance.

12. Machine Learning Models

No.	Machine Learning Model	Short Description (~20 words)
1	Logistic Regression Classification	A linear classification algorithm that predicts the probability of binary outcomes using the sigmoid function for accurate decision-making.
2	Decision Tree Classification	A tree-based algorithm that classifies data by splitting features into decision rules, producing simple, interpretable, and accurate predictions.
3	Random Forest Classification	An ensemble learning method combining multiple decision trees to improve prediction accuracy, reduce overfitting, and increase model robustness.
4	Gradient Boosting Classification	A boosting algorithm that sequentially builds decision trees, correcting previous errors to achieve highly accurate classification performance.
5	XG Boost Classification	An optimized gradient boosting algorithm delivering faster training, higher accuracy, regularization, and excellent performance on structured datasets.
6	K-Nearest (KNN) Classification	A distance-based algorithm that classifies data by identifying the majority class among the nearest neighbouring observations.
7	Gaussian Naive Bayes (GNB) Classification	A probabilistic classifier assuming independent features with Gaussian distribution, enabling fast, efficient, and reliable predictions on continuous data.
8	Support Vector Classification (SVC)	A powerful algorithm that identifies the optimal hyperplane separating classes while maximizing margins for robust and accurate classification.

Decision Tree Plot



13. Model Training

<i>Topic</i>	<i>Short Description (≈20 words)</i>
Training Process	The machine learning model learns patterns from training data by adjusting parameters to accurately predict fraudulent and legitimate transactions.
Hyperparameter Tuning	Hyperparameter tuning optimizes model settings to improve prediction accuracy, reduce overfitting, and achieve better overall performance on unseen data.
Cross-Validation	Cross-validation evaluates model performance by splitting data into multiple folds, ensuring reliable, consistent, and unbiased prediction results across datasets.

14. Model Evaluation

<i>Evaluation Metric</i>	<i>Short Description (≈20 Words)</i>
Accuracy	Measures the overall percentage of correctly classified transactions by comparing all correct predictions with the total number of predictions.
Precision	Indicates how many transactions predicted as fraudulent were actually fraudulent, minimizing false positive predictions and unnecessary investigations.
Recall	Measures the model's ability to correctly identify actual fraudulent transactions while minimizing missed fraud cases (false negatives).
F1 Score	Combines precision and recall into a single metric, providing a balanced evaluation for datasets with class imbalance.
ROC-AUC Score	Evaluates the model's ability to distinguish fraudulent from legitimate transactions across different classification threshold values effectively.
Confusion Matrix	Displays actual versus predicted classifications using true positives, true negatives, false positives, and false negatives for performance analysis.
Classification Report	Summarizes model performance using precision, recall, F1-score, and support for each class in a comprehensive evaluation report.

15. Model Comparison

```
[630]: summary_1
```

```
[630]:
```

	Model Name	Accuracy Score	Precision Score	Recall Score	F1 Score
0	Logistics Regression	90.86	87.85	93.91	90.78
1	Decision Tree Classification	90.92	85.54	97.52	91.14
2	Random Forest Classification	90.92	86.84	95.52	90.97
3	Gradient Boosting Classification	90.92	85.54	97.52	91.14
4	XGB Classification	90.92	86.84	95.52	90.97
5	KNN Classification	89.56	86.81	92.22	89.43
6	GNB Classification	89.10	84.14	95.19	89.33
7	Support Vector Classification	90.99	86.09	96.85	91.15

```
[644]: summary_2
```

```
[644]:
```

	Model Name	True Negative	False Positive	False Negative	True Positive
0	Logistics Regression	9164	1243	583	8990
1	Decision Tree Classification	8829	1578	237	9336
2	Random Forest Classification	9021	1386	429	9144
3	Gradient Boosting Classification	8829	1578	237	9336
4	XGB Classification	9021	1386	429	9144
5	KNN Classification	9066	1341	745	8828
6	GNB Classification	8689	1718	460	9113
7	Support Vector Classification	8909	1498	302	9271

```
[668]: summary_3
```

```
[668]:
```

	index	ROC AUC Score
0	Logistics Regression	96.34
1	Decision Tree Classification	95.22
2	Random Forest Classification	96.19
3	Gradient Boosting Classification	95.93
4	XGB Classification	96.19
5	KNN Classification	94.52
6	GNB Classification	95.06
7	Support Vector Classification	No Need

16. Best Model Selection

After evaluating multiple machine learning algorithms, **XG Boost Classification** was selected as the final model for the fraud detection system due to its superior overall performance, robustness, and suitability for real-world deployment. The model achieved an **Accuracy of 90.92%**, **Precision of 86.84%**, **Recall of 95.52%**, and an **F1 Score of 90.97%**, demonstrating an excellent balance between accurately identifying fraudulent transactions and minimizing false positives.

The confusion matrix further validated the model's effectiveness, correctly classifying **9,144 fraudulent transactions (True Positives)** and **9,021 legitimate transactions (True Negatives)**, while producing **1,386 False Positives** and **429 False Negatives**. The low number of false negatives is particularly significant in fraud detection, as it indicates that the model successfully identified the majority of fraudulent transactions, thereby reducing potential financial losses.

Although the **Gradient Boosting** model achieved a marginally higher Recall, **XG Boost** was selected because it offers several practical advantages for enterprise applications. These include built-in **L1 and L2 regularization** to reduce overfitting, **parallel processing** for faster training, efficient handling of **missing values**, automatic optimization of tree construction, and strong generalization performance on large and complex datasets. Additionally, XG Boost consistently delivers high predictive accuracy while maintaining computational efficiency, making it well suited for large-scale fraud detection systems.

Given its excellent predictive performance, scalability, and reliability, **XG Boost Classification** is the most appropriate model for this project. Its ability to accurately detect fraudulent transactions while maintaining a low error rate makes it an ideal choice for deployment in banking, fintech, retail, and digital payment platforms, where real-time fraud detection is critical for minimizing financial risk and enhancing customer trust.

```
[675]: XGB_Model
```

```
[675]: XGBClassifier
XGBClassifier(base_score=None, booster=None, callbacks=None,
              colsample_bylevel=None, colsample_bynode=None,
              colsample_bytree=None, device=None, early_stopping_rounds=None,
              enable_categorical=False, eval_metric=None, feature_types=None,
              gamma=None, grow_policy=None, importance_type=None,
              interaction_constraints=None, learning_rate=None, max_bin=None,
              max_cat_threshold=None, max_cat_to_onehot=None,
              max_delta_step=None, max_depth=None, max_leaves=None,
              min_child_weight=None, missing=nan, monotone_constraints=None,
              multi_strategy=None, n_estimators=None, n_jobs=None,
              num_parallel_tree=None, random_state=None, ...)
```

17. Prediction Results

The machine learning model was evaluated using a test dataset of **100 transaction records**. The model generated prediction probabilities for each transaction, allowing fraud risk to be assessed with confidence scores. A comparison of the **actual** and **predicted** classes demonstrated strong classification performance. The model achieved an **Accuracy of 87%, Precision of 83%, Recall of 91%, and an F1-Score of 87%**. The confusion matrix showed **42 True Negatives, 45 True Positives, 9 False Positives, and 4 False Negatives**. These results indicate that the model is highly effective at identifying fraudulent transactions while maintaining a good balance between detecting fraud and minimizing incorrect classifications, making it suitable for fraud risk prediction.

18. Business Insights

The analysis identified several patterns associated with fraudulent transactions. High-risk activities were commonly linked to unusually large transaction amounts, frequent transactions within a short period, international payments, transactions from unfamiliar locations, and the use of high-risk devices. Feature importance analysis showed that variables such as transaction amount, transaction frequency, unusual location, account age, previous fraud history, failed transaction attempts, and device risk were the strongest predictors of fraud. Based on these insights, businesses should implement real-time fraud detection, strengthen authentication for high-risk transactions, continuously monitor suspicious behavior, and regularly update machine learning models to reduce financial losses while improving customer trust and payment security.

19. Saved The Model

Saved the trained machine learning model using Joblib for efficient serialization and deployment, enabling fast loading, reusable predictions, seamless integration into applications, and consistent performance without retraining the model each time it is used.

20. Limitations

This project has several limitations. The dataset is simulated and may not capture all real-world fraud patterns, seasonal trends, or emerging fraud techniques, which can affect the model's ability to generalize. The machine learning model assumes that historical transaction patterns are representative of future behavior and that the selected features are sufficient to distinguish fraudulent from legitimate transactions. Additionally, the dataset may contain class imbalance, making fraud detection more challenging. The model's performance also depends on data quality, feature engineering, and regular retraining. Real-time deployment may require greater computational resources, continuous monitoring, and integration with production systems to maintain accuracy and reliability.

21. Future Enhancements

Future enhancements for this fraud detection system focus on improving accuracy, scalability, and real-world applicability. Collecting larger and more diverse transaction datasets can help the model learn evolving fraud patterns and improve its generalization. Advanced feature engineering techniques can uncover hidden relationships and enhance predictive performance. Deep learning models, such as Artificial Neural Networks (ANNs) and Long Short-Term Memory (LSTM) networks, can be explored to capture complex transaction behaviors. The system can also be extended to support real-time fraud detection using streaming data. Finally, deploying the solution on cloud platforms such as Google Cloud, AWS, or Microsoft Azure would enable scalable, secure, and production-ready fraud detection services.

22. Conclusion

This project successfully demonstrates how machine learning can be applied to detect fraudulent transactions in retail and e-commerce payment systems. By analyzing historical transaction data, performing data preprocessing, conducting exploratory data analysis, and engineering meaningful features, the project identified patterns associated with fraudulent behavior. Multiple classification models were developed and evaluated using performance metrics such as Accuracy, Precision, Recall, F1-Score, and ROC-AUC. The best-performing model provided reliable predictions while effectively balancing fraud detection with minimizing false positives. The findings highlight that factors such as unusual transaction amounts, high transaction frequency, unfamiliar locations, risky devices, and previous fraud history are strong indicators of fraudulent activity. The proposed solution can support businesses in identifying high-risk transactions, reducing financial losses, and improving customer trust. Although the model performs well, continuous retraining with updated data and integration with real-time monitoring systems can further enhance its effectiveness, making it a valuable decision-support tool for modern fraud prevention.

23. References

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